

Phylogenetic Analysis Using NCBI and Galaxy

GROUP 4 - Anatidae
BIO 300 - Cell & Molecular Biology

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Introduction

The cytochrome c oxidase subunit I (COI) gene is a mitochondrial gene commonly used to identify species and study their evolutionary relationships. It contains enough genetic differences to distinguish closely related species while remaining similar enough to compare different groups of organisms. In this activity, COI sequences from selected species in the family Anatidae were used to construct a phylogenetic tree, with one species outside the family included as an outgroup to root the tree. The phylogenetic tree shows the evolutionary relationships and genetic similarities among the selected duck species.

Table 1. *Anatidae species, NCBI accession number, COI sequence length, Record title, and outgroup designation.*

Species	NCBI accession number	Sequence Length	Record Title	Focal taxon / Outgroup
<i>Phasianus colchicus</i> (Ring-necked pheasant)	MK262683.1	648 bp	Phasianus colchicus voucher SP524 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial	Outgroup
<i>Lophodytes cucullatus</i> (Hooded merganser)	AY666346.1	590 bp	Lophodytes cucullatus voucher 1B-121 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial	Focal taxon
<i>Mergus merganser</i> (Common merganser)	GU571975.1	648 bp	Mergus merganser voucher BISE-Aves164 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial	Focal taxon
<i>Anas rubripes</i> (American black duck)	AY666221.1	692 bp	Anas rubripes voucher 1B-726 cytochrome oxidase subunit	Focal taxon

			1 (COI) gene, partial cds; mitochondrial	
<i>Anas platyrhynchos</i> (Mallard duck)	MK262361.1	648 bp	Anas platyrhynchos voucher SP331 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial	Focal taxon
<i>Anas clypeata</i> (Northern shoveler)	JF499085.1	675 bp	Anas clypeata voucher USNM:641832 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial	Focal taxon
<i>Anas strepera</i> (Gadwall duck)	GU571726.1	648 bp	Anas strepera voucher BISE-Aves8 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial	Focal taxon

Methods

COI gene sequences of the selected Anatidae species were retrieved from the NCBI database, together with *Phasianus colchicus* as the outgroup. All sequences were downloaded in FASTA format and imported into Galaxy for analysis. Multiple sequence alignment was performed using MAFFT to examine the genetic similarities and differences among the sequences. Following alignment, a consensus sequence was produced to summarize the conserved nucleotide positions shared among the species. A phylogenetic tree was then generated using FastTree to visualize the evolutionary relationships, with *Phasianus colchicus* used as the outgroup to root the phylogenetic tree.

The figures below illustrate the workflow used to construct the phylogenetic tree from the COI sequences.

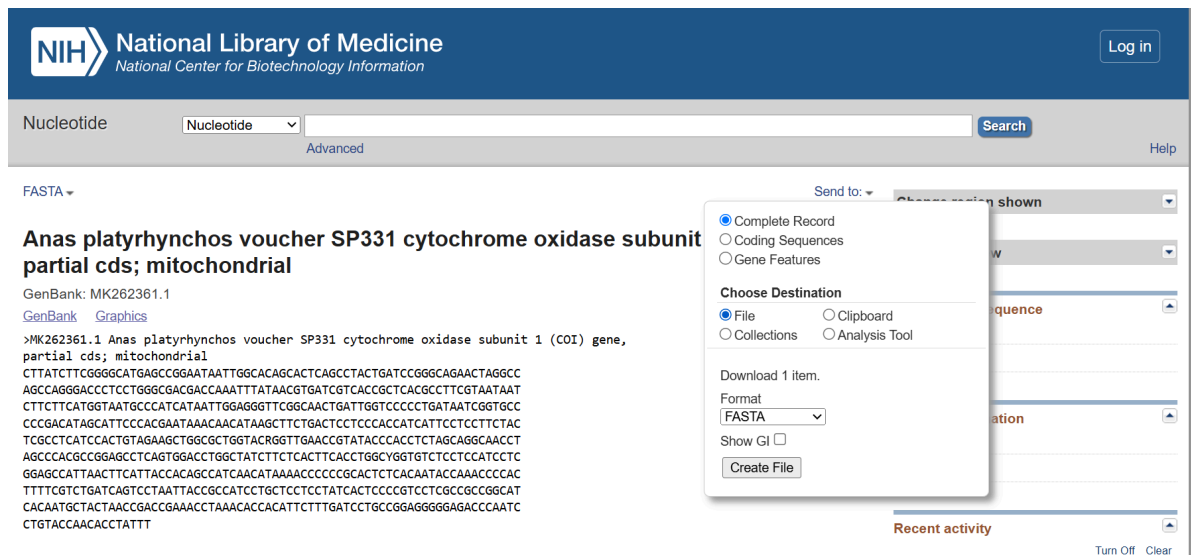


Figure 1. Workflow of how sequence data was collected from NCBI Nucleotide. Searching for each species using its scientific name and the COI/COX1 gene. Checking each sequence to make sure it covers the correct region and length. Saving the original, unchanged FASTA files with clear names for analysis.

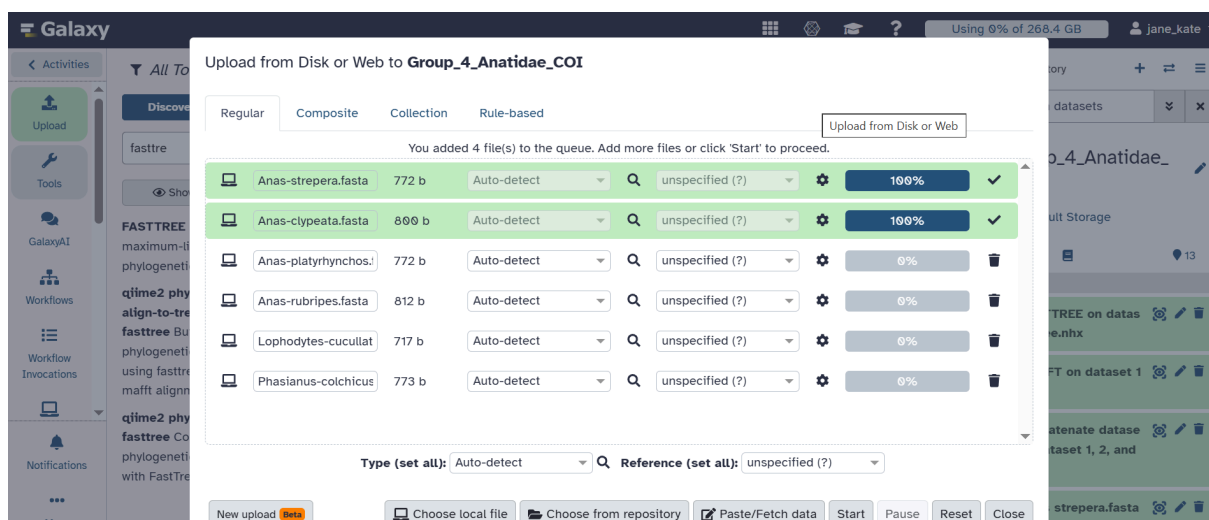


Figure 2. Uploading and preparing FASTA files in Galaxy. Downloaded COI sequences from NCBI were uploaded into the Galaxy history. Each file was checked to confirm the correct FASTA format and renamed with clear species labels for analysis.

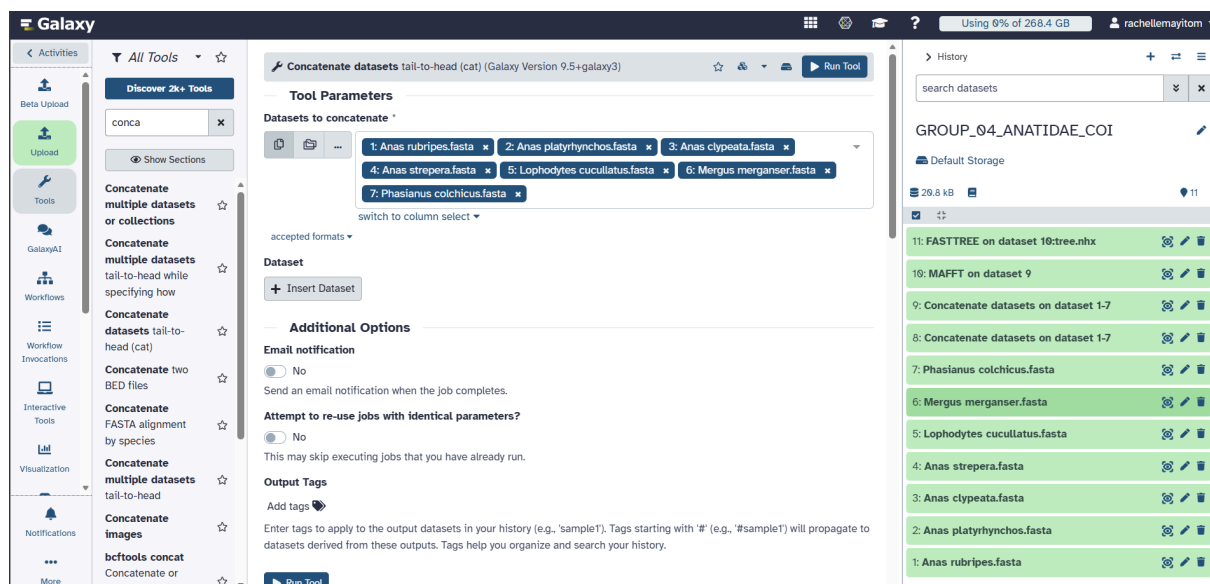


Figure 3. Combining sequences into a multi-FASTA file in Galaxy. Concatenating individual COI FASTA records into a single dataset.

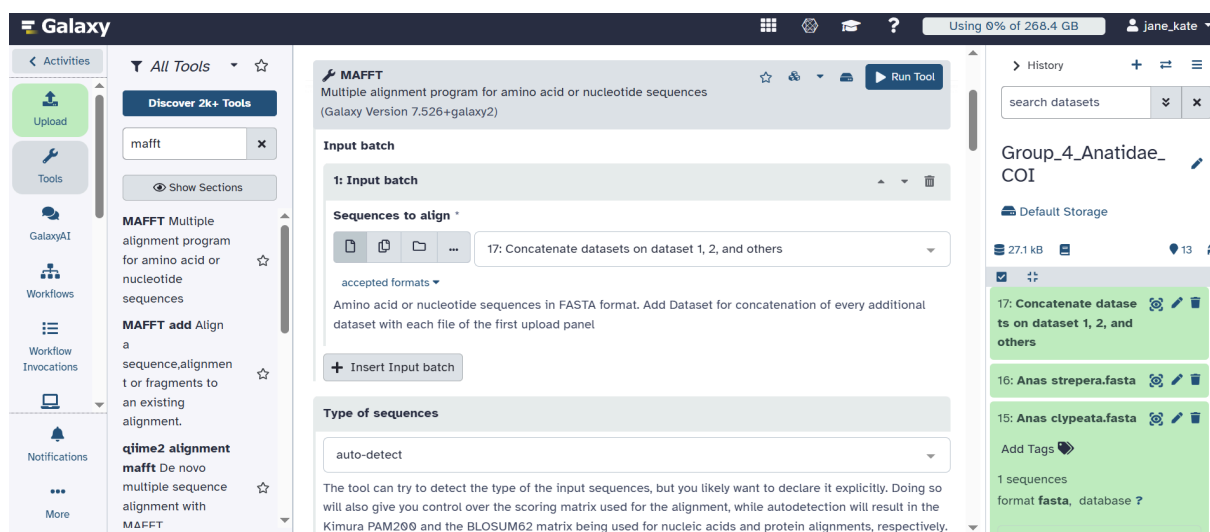


Figure 4. Running the MAFFT multiple sequence alignment tool in Galaxy. The selected COI sequences were aligned using MAFFT to identify conserved and variable nucleotide positions among the species.

	Id	Actual Id	Begin	End	Link	Real Length	Aligned Leng...	Left
1	P.colchicus	P.colchicus	1	648		648	648	5
2	M.merganser	M.merganser	1	648		648	648	5
3	L.cucullatus	L.cucullatus	1	590		590	590	102
4	A.strepera	A.strepera	1	648		648	648	5
5	A.clypeata	A.clypeata	1	675		675	675	17
6	A.platyrhynchos	A.platyrhynchos	1	648		648	648	5
7	A.rubripes	A.rubripes	1	692		692	692	0

Figure 5. Visualization of the MAFFT multiple sequence alignment. Alignment results and sequence logos were generated in Galaxy to inspect conserved regions, variations, and sequence similarities across all included species.

The screenshot displays the Galaxy web interface for the FASTTREE tool. The tool parameters are configured as follows:

- Tool Parameters:** FASTTREE build maximum-likelihood phylogenetic trees (Galaxy Version 2.1.19+galaxy1)
- Aligned sequences file (FASTA or Phylip format):** fasta
- FASTA file:** 18: MAFFT on dataset 17
- Accepted formats:** dropdown menu
- Allow spaces and other restricted characters (but not ' ') in sequence and quote names in the output tree:** No
- Set starting tree(s):** No starting trees

The history panel on the right shows a list of datasets:

- 18: MAFFT on dataset 17
- 17: Concatenate datasets on dataset 1, 2, and others
- 16: Anas strepera.fasta
- 15: Anas clypeata.fasta

Figure 6. Construction of the phylogenetic tree using FastTree in Galaxy. The aligned COI sequences were analyzed using the maximum-likelihood method to infer the evolutionary relationships among the selected species.

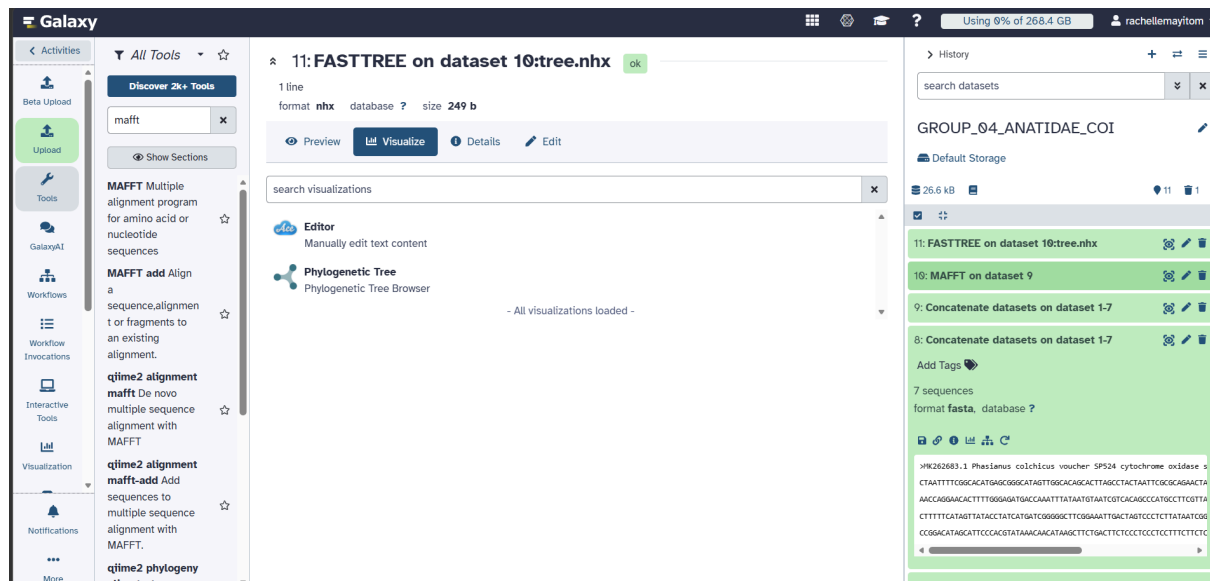


Figure 7. Visualization of the phylogenetic tree generated by FastTree in Galaxy, showing the branching pattern and evolutionary relationships among the aligned COI sequences.

Summary of Conserved and Variable Sites

The consensus sequence generated from the aligned COI sequences contained a total of 692 nucleotide positions, consisting of 507 conserved sites (73%) and 185 variable sites (27%). The high number of conserved sites indicates that the COI gene is highly similar among the selected Anatidae species, reflecting its important biological function and evolutionary conservation. In contrast, the variable sites were represented by IUPAC ambiguity codes (e.g., R, Y, W, M, and K), indicating differences in nucleotides among the species. These genetic variations provide useful information for distinguishing closely related species and support the construction of the phylogenetic tree by revealing their evolutionary relationships.

The phylogenetic tree constructed from the COI sequences shows the evolutionary relationships among the selected Anatidae species, with *Phasianus colchicus* serving as the outgroup to root the tree. As expected, the outgroup is separated from all Anatidae species, indicating that it is more distantly related and providing a reference for the direction of evolutionary divergence. Within the Anatidae group, *Mergus merganser* and *Lophodytes cucullatus* cluster together, indicating that they share a more recent common ancestor and are more closely related to each other than to the other species. Among the *Anas* species, *Anas platyrhynchos* and *Anas rubripes* form the closest pair, suggesting the highest genetic similarity based on their COI sequences. *Anas strepera* is closely related to this pair, while *Anas chrypeata* branches earlier, indicating that it is more distantly related within the *Anas* group. Overall, the tree shows that species belonging to the same genus cluster together, demonstrating that the COI gene is effective for distinguishing closely related species and inferring their evolutionary relationships.

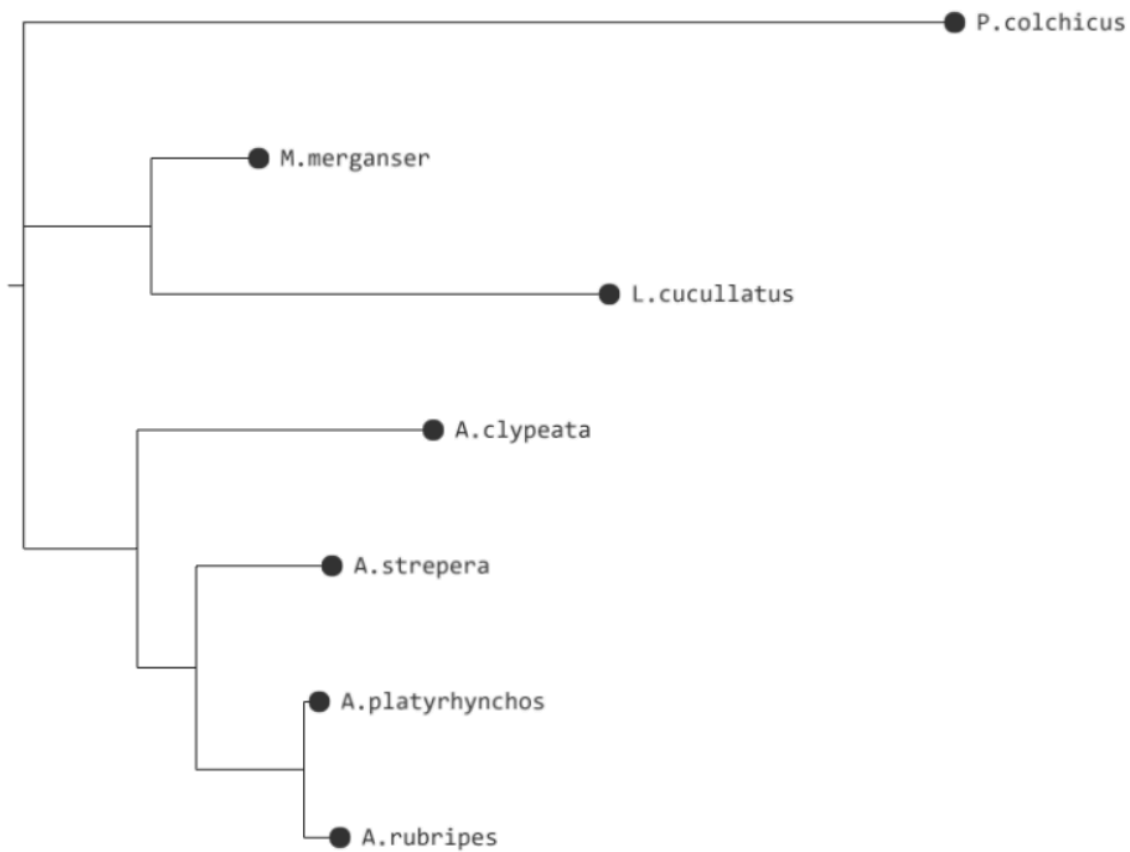


Figure 8. *Phylogenetic tree of the selected Anatidae species based on mitochondrial COI sequences generated using FastTree.*

Comparison with Published Studies

The phylogenetic tree generated from the COI sequences is generally consistent with recent studies on the evolutionary relationships within the family Anatidae. In the present study, *Anas platyrhynchos* and *Anas rubripes* formed the closest clade, indicating a high degree of genetic similarity. This agrees with the findings of Brown et al. (2022), who reported that closely related duck species share a recent common ancestry and exhibit strong genetic similarity despite occasional hybridization.

The tree also grouped *Mergus merganser* and *Lophodytes cucullatus* into the same clade, separate from the *Anas* species, reflecting their close evolutionary relationship within the merganser lineage. Similarly, Deef (2021) found that mitochondrial COI sequences effectively distinguished closely related duck species and produced phylogenetic groupings that corresponded with their evolutionary relationships.

Furthermore, the clear separation of *Phasianus colchicus* from all Anatidae species confirms its suitability as an outgroup for rooting the phylogenetic tree and determining the

direction of evolutionary divergence. Overall, the topology obtained in this study agrees with previous molecular studies, supporting the use of the mitochondrial COI gene as a reliable marker for species identification and for reconstructing phylogenetic relationships among ducks.

Although the phylogenetic relationships observed in this study are consistent with previous molecular studies, it is important to recognize that using the COI gene alone has some limitations. As a single mitochondrial marker, COI reflects only maternal inheritance and may not fully represent the complete evolutionary history of species, particularly in groups where hybridization is common, such as ducks (Toews & Brelsford, 2012). Nevertheless, COI remains a reliable marker for species identification and phylogenetic analysis, and combining it with additional mitochondrial or nuclear genes could further improve the accuracy and resolution of evolutionary relationships (Hebert et al., 2003).

References

- Brown, J. I., Hernández, F., Engilis, A., Jr., Hernández-Baños, B. E., Collins, D., Lavretsky, P., et al. (2022). *Genomic and morphological data shed light on the complexities of shared ancestry between closely related duck species*. *Scientific Reports*, 12, 10212. <https://doi.org/10.1038/s41598-022-14270-2>
- Deef, L. E. M. (2021). *Phylogenetic relationships among puddle duck species in Egypt using COI gene variations in mtDNA*. *Russian Journal of Genetics*, 57(10), 1189–1197. <https://doi.org/10.1134/S1022795421100045>
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- Toews, D. P. L., & Brelsford, A. (2012). The biogeography of mitochondrial and nuclear discordance in animals. *Molecular Ecology*, 21(16), 3907–3930. <https://doi.org/10.1111/j.1365-294X.2012.05664.x>